

AI Interviewer Proposal

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Research Objectives

The primary objective of this project is to design a comprehensive AI Interview Assessment System capable of evaluating candidate performance through multimodal analysis of video, audio, and textual information.

The specific objectives are:

1. To develop a multimodal interview assessment framework that integrates video, audio, and transcript-based features.
2. To improve interview integrity through candidate verification and suspicious behaviour detection mechanisms.
3. To investigate the effectiveness of feature fusion techniques in combining multimodal interview signals for candidate evaluation.
4. To design an explainable and fair AI assessment system that reduces bias and improves transparency in recruitment decisions.
5. To provide recruiters with objective, data-driven insights while maintaining human oversight in the final hiring decision process.

Research Questions

RQ1:

Can multimodal analysis of video, audio, and transcript information improve interview assessment accuracy compared to transcript-only evaluation?

RQ2:

Can suspicious behaviour detection improve interview integrity and reduce the risk of unfair candidate assessment?

RQ3:

Does multimodal feature fusion outperform independent modality scoring in predicting candidate performance and hireability?

RQ4:

Can fairness monitoring and explainable AI improve trustworthiness and transparency in AI-assisted recruitment systems?

RQ5:

How closely do AI-generated interview assessments align with evaluations performed by human recruiters?

1.0 System Overview

The proposed AI Interview Assessment System is an end-to-end multimodal evaluation framework designed to assess candidate performance objectively while maintaining interview integrity and reducing human bias. The system analyzes visual behavior, speech characteristics, interview content, candidate authenticity, and suspicious activities before generating explainable assessment scores. Unlike traditional interview evaluation systems that focus only on interview answers, this framework incorporates quality assessment, candidate verification, suspicious behavior detection, feature fusion, fairness monitoring, and explainable AI to improve reliability and trustworthiness.

2.0 Module Explanation

2.0.1 RecruitView Dataset and Annotation Strategy

The proposed system is partially inspired by the RecruitView dataset and benchmark. Understanding how RecruitView labels were generated is important because these labels serve as the ground truth for model training and evaluation.

According to the RecruitView paper, candidate interview videos were manually evaluated and annotated across multiple dimensions related to interview performance and personality characteristics. Human annotators assessed factors such as communication ability, confidence, professionalism, and overall interview performance. Multiple annotation dimensions were used to capture different aspects of candidate behaviour rather than relying on a single interview score.

The paper also reports that most quantitative labels follow approximately normal distributions, indicating that the dataset contains candidates across different performance levels rather than being dominated by only high-performing or low-performing candidates. This improves the suitability of the dataset for machine learning model development.

The use of human annotation is particularly important because interview quality is inherently subjective. By collecting labels from multiple evaluators and establishing consistent scoring criteria, the dataset provides a more reliable benchmark for training and evaluating AI interview assessment systems.

The proposed system adopts a similar philosophy by combining multiple behavioural indicators and using human evaluation as part of the model validation process.

2.1 Quality Assessment Module

Purpose

The Quality Assessment Module serves as the first filtering stage of the pipeline. Before evaluating a candidate, the system must ensure that the interview recording meets minimum quality requirements. Poor-quality videos may produce inaccurate feature extraction and unfair assessments.

Libraries

- **OpenCV**
- **MediaPipe**
- **Librosa**
- **FFmpeg**

Features Evaluated

Face Visibility Check

The system calculates the percentage of video frames where a face is successfully detected.

Example:

- Face visible in 95% of frames → Valid
- Face visible in 20% of frames → Poor quality

Audio Quality Check

The system evaluates:

- Background noise level
- Signal-to-noise ratio
- Audio volume consistency

Lighting Quality Check

The system evaluates:

- Brightness
- Contrast

- Visibility of facial features

Video Duration Check

The system verifies whether sufficient interview content exists.

Example:

- Duration > 30 seconds → Acceptable
- Duration < 10 seconds → Insufficient evidence

Importance

Poor-quality recordings can cause incorrect emotion detection, inaccurate speech transcription, and unreliable assessment results. Therefore, quality control is necessary before proceeding to further analysis.

2.2 Candidate Verification Module

Purpose

The Candidate Verification Module ensures that the person being evaluated is the intended candidate and that no identity-related issues occur during the interview.

Libraries

- MediaPipe Face Detection
- DeepFace
- face_recognition
- OpenCV

Features Evaluated

Face Count Detection

The system counts the number of visible faces.

Example:

- One face → Valid
- Two or more faces → Suspicious

Identity Verification

The system compares facial embeddings throughout the interview.

Example:

- Same face detected throughout interview → Valid
- Different face detected later → Potential identity switch

Candidate Presence Tracking

The system monitors how often the candidate leaves the camera view.

Importance

Online interviews are vulnerable to identity fraud, proxy candidates, and external assistance. This module helps maintain assessment integrity before candidate evaluation begins.

2.3 Video Feature Extraction Module

Purpose

This module extracts non-verbal behavioral information from interview videos.

Libraries

- MediaPipe FaceMesh
- MediaPipe Pose
- DeepFace
- OpenCV

Features Extracted

Eye Contact Score

Measures candidate attention and engagement.

Metrics:

- Eye contact ratio
- Average gaze duration

Facial Emotion Score

Detects emotional states.

Examples:

- Happy
- Neutral
- Confident
- Nervous

Head Pose Analysis

Measures:

- Looking away frequency
- Head stability

Posture Analysis

Measures:

- Upright posture
- Slouching
- Leaning

Gesture Stability

Measures:

- Hand movement variance
- Body movement variance

Importance

Human interviewers naturally evaluate body language. This module quantifies those observations and reduces subjectivity.

2.4 Audio Feature Extraction Module

Purpose

This module analyzes how candidates speak rather than what they say.

Libraries

- Librosa
- PyDub
- Parselmouth
- SpeechBrain
- spaCy

Features Extracted

Speech Rate

Words spoken per minute.

Pause Ratio

Percentage of interview time spent in silence.

Pitch Stability

Measures consistency of vocal pitch.

Voice Jitter

Measures micro-instability in speech.

Voice Emotion

Detects:

- Confidence
- Stress
- Excitement

Filler Word Frequency

Detects:

- Um
- Uh
- Like
- Basically

Importance

Communication skills are essential in interviews. Audio analysis provides objective measurements that complement visual observations.

2.5 Speech-to-Text Module

Purpose

Converts spoken interview responses into text.

Libraries

- Whisper Large
- WhisperX

Outputs

- Transcript
- Word timestamps
- Transcription confidence

Importance

Language models require textual input to analyze response quality. Accurate transcription is therefore critical for downstream evaluation.

2.6 LLM-Based Content Evaluation Module

Purpose

Evaluates the semantic quality of interview responses.

Libraries

- Llama 3
- Gemma
- Sentence-BERT
- spaCy

Features Evaluated

Answer Relevance

Measures how well the response addresses the interview question.

STAR Framework Analysis

Detects:

- Situation
- Task
- Action
- Result

Communication Quality

Measures:

- Clarity
- Structure
- Coherence

Technical Knowledge

Evaluates:

- Correctness

- Depth of understanding

Vocabulary Richness

Measures:

- Lexical diversity
- Language complexity

Importance

Candidates should not be evaluated solely based on confidence or appearance. The actual quality of their answers remains one of the most important hiring criteria.

2.7 Suspicious Behavior Detection Module

Purpose

Most AI interview systems focus on evaluating candidate performance while assuming that candidates behave honestly during the interview. However, in real-world online interviews, candidates may receive external assistance, read prepared scripts, switch identities, or use secondary devices to gain an unfair advantage. These behaviors can significantly affect the reliability of interview assessments.

The purpose of the Suspicious Behavior Detection Module is not to automatically reject candidates, but to identify unusual patterns and generate a Suspicion Score that can be reviewed by recruiters.

System Workflow

This module receives inputs from both the video and audio streams after preprocessing.

Step 1: Multiple Person Detection

Using MediaPipe Face Detection or YOLO, the system continuously monitors the number of faces present in each frame.

Example:

Frame 1:

- Face Count = 1

Frame 300:

- Face Count = 2

Frame 500:

- Face Count = 2

This may indicate:

- External assistance
- Another person entering the interview
- Shared interview environment

Instead of immediately marking the interview as invalid, the system records:

- Number of occurrences
- Duration of appearance
- Percentage of interview affected

These values become features used later by the fusion module.

Step 2: Identity Consistency Verification

DeepFace or face_recognition extracts facial embeddings at different timestamps.

Example:

- Beginning of interview
- Middle of interview
- End of interview

The system compares embedding similarities.

Possible outcomes:

Identity Match = 98%

→ Same person

Identity Match = 45%

→ Possible identity switch

This prevents proxy interviews where one candidate starts the interview and another candidate completes it.

Step 3: Eye Gaze Monitoring

MediaPipe FaceMesh is used to estimate eye gaze direction and head orientation.

Measurements:

- Looking at camera percentage
- Looking away percentage
- Continuous off-screen duration

Example:

Candidate spends:

- 30% looking at camera
- 70% looking left

Possible explanation:

- Reading notes
- Looking at a second monitor
- Using ChatGPT assistance

The system records these behaviors rather than making assumptions.

Step 4: Phone and Device Detection

YOLO object detection can identify:

- Smartphones
- Tablets
- Additional screens

The system records:

- Device type
- Detection frequency
- Duration of visibility

Example:

Phone detected for 90 seconds during the interview.

This becomes another suspicious behavior indicator.

Step 5: Script Reading Detection

This feature combines multiple signals:

Video signals:

- Repetitive eye movement
- Consistent head position
- Limited facial variation

Audio signals:

- Unusually smooth delivery
- Very low pause ratio
- Extremely consistent speech speed

Transcript signals:

- Perfectly structured responses
- Very formal language
- Low spontaneity

Instead of relying on a single feature, the system combines evidence from multiple modalities.

Step 6: Suspicion Score Generation

The suspicious behaviors are converted into numerical features.

Example:

Multiple Face Risk = 0.6

Identity Risk = 0.1

Looking Away Risk = 0.8

Phone Usage Risk = 0.5

Script Reading Risk = 0.7

The system computes:

Suspicion Score =

$w_1 \times \text{Multiple Face Risk}$

+

$w_2 \times \text{Identity Risk}$

+

$w_3 \times \text{Looking Away Risk}$

+

$w_4 \times \text{Phone Usage Risk}$

+

$w_5 \times \text{Script Reading Risk}$

where

$\sum w_i = 1$

Output:

- Low Risk (0-30)
- Medium Risk (31-60)
- High Risk (61-100)

Why This Module Is Important

Most AI interview systems only evaluate performance. This module evaluates interview integrity. A candidate with excellent answers but a high suspicion score may require manual

review before a hiring decision is made. This improves trustworthiness and reduces the possibility of cheating during online interviews.

2.8 Feature Fusion Module

Purpose

The Feature Fusion Module is the central component of the entire AI Interview Assessment System.

Many systems simply calculate independent scores and average them together.

For example:

Final Score =
 $(\text{Video Score} + \text{Audio Score} + \text{Content Score}) / 3$

This approach ignores the relationships between different features.

The Feature Fusion Module instead combines all extracted features into a unified representation before prediction.

Why Fusion Is Necessary

Consider two candidates.

Candidate A:

- Excellent eye contact
- Strong posture
- Weak technical answers

Candidate B:

- Nervous body language
- Excellent technical knowledge

If simple averaging is used, important information may be lost.

Human interviewers naturally combine multiple observations simultaneously.

The Feature Fusion Module attempts to replicate this process.

Step 1: Feature Collection

Features are collected from all previous modules.

Video Features

1. Eye Contact Ratio
2. Facial Emotion Score
3. Posture Stability
4. Gesture Stability
5. Head Movement Score

Audio Features

1. Speech Rate
2. Pause Ratio
3. Pitch Stability
4. Voice Jitter
5. Emotion Confidence
6. Filler Word Frequency

Content Features

1. Answer Relevance
2. STAR Score
3. Technical Knowledge Score
4. Communication Score
5. Vocabulary Richness

Integrity Features

1. Suspicion Score
2. Identity Verification Score
3. Interview Quality Score

Step 2: Feature Normalization

Different features have different scales.

Example:

Speech Rate = 135

Eye Contact = 0.82

Suspicion Score = 75

These values cannot be directly combined.

Therefore, Min-Max Scaling or Standardization is applied.

Example:

Speech Rate $\rightarrow$ 0.75

Eye Contact $\rightarrow$ 0.82

Suspicion Score $\rightarrow$ 0.75

Now all features share the same scale.

Step 3: Feature Engineering

Additional high-level features can be generated.

Example:

Confidence Score =

- Eye Contact
- Speech Rate
- Pause Ratio
- Voice Jitter

Professionalism Score =

- Posture Stability
- Emotion Stability
- Communication Quality

Communication Score =

- Speech Rate
- Vocabulary Richness
- STAR Score

This creates more meaningful representations than raw features alone.

Step 4: Unified Feature Vector Construction

All normalized features are combined into a single vector.

Example:

```
[  
Eye Contact,  
Posture Stability,  
Speech Rate,  
Pause Ratio,  
STAR Score,  
Technical Score,  
Suspicion Score  
]
```

This vector becomes the input to the final machine learning model.

Step 5: Model Learning

The scoring model (XGBoost, LightGBM, Random Forest) learns patterns across all modalities simultaneously.

Example:

The model may discover:

High Eye Contact

+

Low Pause Ratio

+

High STAR Score

→ Strong Candidate

Or:

Excellent Content

+

Very High Suspicion Score

→ Requires Human Review

This interaction between features would not be captured by simple averaging.

Step 6: Multi-Level Score Generation

Rather than generating only one score, the fusion layer can produce hierarchical assessments.

Confidence Score

Uses:

- Eye Contact
- Speech Rate
- Voice Stability

Professionalism Score

Uses:

- Posture
- Gesture Stability
- Facial Emotion

Communication Score

Uses:

- Speech Rate
- STAR Score
- Vocabulary Richness

Technical Score

Uses:

- Technical Knowledge
- Relevance

Hireability Score

Uses:

- All previous scores
- Suspicion Score

- Interview Quality Score

Why This Module Is Important

This module is the decision-making core of the system. Without feature fusion, the system would evaluate video, audio, and transcript information separately. With feature fusion, the system can understand how different behaviors interact and contribute to interview performance. This allows the AI Interview Assessment System to behave more similarly to a human interviewer, who naturally combines visual observations, communication skills, answer quality, and interview integrity when making hiring decisions.

2.9 Interview Scoring Module

Purpose

Transforms extracted features into meaningful assessment scores.

Libraries

- XGBoost
- LightGBM
- Random Forest
- Scikit-Learn

Outputs

Confidence Score

Based on:

- Eye contact
- Speech rate
- Pause ratio
- Voice stability

Fluency Score

Based on:

- Filler words
- Speech rate
- Grammar quality

Professionalism Score

Based on:

- Posture
- Emotion
- Communication behavior

Answer Quality Score

Based on:

- Relevance
- STAR framework
- Technical correctness

Hireability Score

Overall candidate suitability.

Example Scoring Formulation

To improve transparency and interpretability, intermediate assessment scores can be calculated using weighted combinations of extracted features.

Confidence Score

Confidence Score =

$0.35 \times \text{Eye Contact Score}$

$+ 0.25 \times \text{Speech Rate Score}$

$+ 0.20 \times \text{Voice Stability Score}$

$+ 0.20 \times \text{Gesture Stability Score}$

Communication Score

Communication Score =

$0.40 \times \text{Answer Relevance Score}$

$+ 0.30 \times \text{STAR Framework Score}$

$+ 0.20 \times \text{Vocabulary Richness Score}$

$+ 0.10 \times \text{Fluency Score}$

Professionalism Score

$$\begin{aligned}\text{Professionalism Score} = & \\ & 0.35 \times \text{Posture Stability Score} \\ & + 0.25 \times \text{Facial Engagement Score} \\ & + 0.20 \times \text{Communication Score} \\ & + 0.20 \times \text{Behaviour Consistency Score}\end{aligned}$$

Technical Competency Score

$$\begin{aligned}\text{Technical Competency Score} = & \\ & 0.50 \times \text{Technical Knowledge Score} \\ & + 0.30 \times \text{Answer Relevance Score} \\ & + 0.20 \times \text{Problem Solving Score}\end{aligned}$$

Hireability Score

$$\begin{aligned}\text{Hireability Score} = & \\ & 0.30 \times \text{Technical Competency Score} \\ & + 0.25 \times \text{Communication Score} \\ & + 0.20 \times \text{Confidence Score} \\ & + 0.15 \times \text{Professionalism Score} \\ & + 0.10 \times \text{Interview Integrity Score}\end{aligned}$$

The weights shown above are illustrative examples and may be optimized through model training and validation experiments.

Final Recommendation Categories

Hireability Score ≥ 80

→ Strongly Recommended

$60 \leq$ Hireability Score < 80

→ Recommended

$40 \leq$ Hireability Score < 60

→ Requires Further Review

Hireability Score < 40

→ Not Recommended

Suspicion Score > 70

→ Manual Review Required

Importance

Provides recruiters with structured, interpretable assessment metrics.

2.10 Fairness and Explainability Module

Purpose

Ensures transparent and unbiased decision-making.

Libraries

- Fairlearn
- IBM AIF360
- SHAP
- LIME

Functions

Bias Detection

Evaluates:

- Gender bias
- Accent bias
- Demographic bias

Explainability

Provides explanations for model decisions.

Example:

Candidate Score = 85

Positive Contributors:

- Strong STAR responses
- High eye contact
- Good technical knowledge

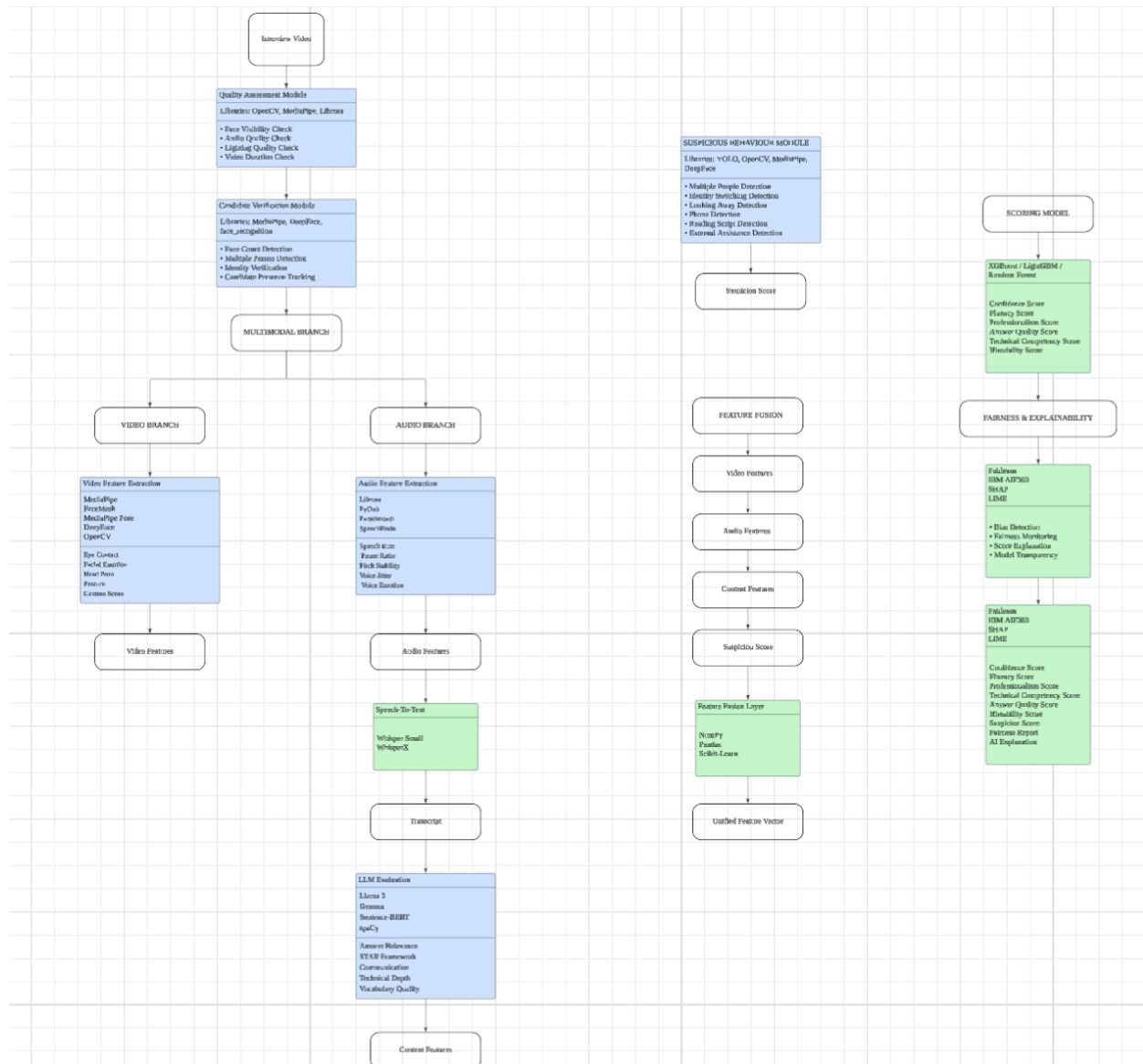
Negative Contributors:

- Frequent pauses
- Low posture stability

Importance

Organizations require trustworthy AI systems. Explainability increases transparency while fairness monitoring reduces ethical and legal risks.

3.0 Flowchart



https://lucid.app/lucidchart/36550e60-618c-4d8a-8bf5-5537aa1607d3/edit?viewport_loc=-2372%2C-2046%2C11958%2C6763%2C0_0&invitationId=inv_9e2446f4-f4f6-4e96-90bf-69a1c33f62a1

4.0 Potential System Failures and Mitigation Strategies

Although the proposed AI Interview Assessment System aims to provide objective and scalable candidate evaluation, several challenges may affect model reliability and fairness. Understanding these limitations is important to ensure responsible deployment.

4.1 Dataset Bias

Problem

The RecruitView dataset primarily contains Indian candidates.

As a result, the model may learn patterns specific to:

- Indian accents
- Indian communication styles
- Cultural speaking behaviors
- Local interview practices

The model may therefore perform poorly when evaluating:

- Malaysian candidates
- Chinese candidates
- Western candidates
- Candidates with different accents

Impact

The model may incorrectly associate certain accents or communication styles with interview quality.

Example:

A highly qualified candidate with a strong regional accent may receive lower scores because the model has limited exposure to similar speech patterns.

Mitigation

Collect Diverse Data

Include candidates from:

- Different countries
- Different ethnic groups
- Different industries

Fairness Monitoring

Use:

- Fairlearn
- IBM AIF360

to continuously monitor bias.

Human Review

Flag uncertain cases for manual evaluation.

4.2 Incorrect Facial Analysis

Problem

Facial expression models may misinterpret emotions.

Example:

A candidate may naturally have a serious facial expression.

The system may incorrectly classify:

Neutral Face → Nervous

or

Focused Face → Unhappy

Impact

Confidence scores may become inaccurate.

Mitigation

Avoid using emotion as the sole indicator.

Instead:

Confidence Score should combine:

- Eye Contact
- Speech Rate
- Voice Stability
- Posture

This reduces dependency on a single feature.

4.3 Eye Contact Misinterpretation

Problem

The system may assume:

Looking away = cheating

However:

The candidate may simply be thinking.

Technical interview candidates often look away while solving problems.

Impact

False suspicion flags.

Mitigation

Use eye gaze as one indicator rather than a direct cheating signal.

Combine with:

- Phone detection
- Multiple face detection
- Script reading indicators

Only flag when multiple suspicious signals occur simultaneously.

4.4 Speech Recognition Errors

Problem

Whisper may incorrectly transcribe:

- Strong accents
- Technical terms
- Industry-specific jargon

Example:

"PyTorch"

may become

"Pie Torch"

Impact

LLM content evaluation becomes inaccurate.

Incorrect transcript → Incorrect answer quality score.

Mitigation

Use:

- WhisperX
- Domain-specific vocabulary correction
- Human review for low-confidence transcripts

Store transcript confidence scores and flag uncertain outputs.

4.5 LLM Hallucination

Problem

Large Language Models may generate incorrect evaluations.

Example:

The candidate provides a correct answer.

The LLM mistakenly interprets it as incomplete.

Impact

Answer quality scores become unreliable.

Mitigation

Use structured prompts.

Example:

Evaluate:

- Relevance
- STAR Framework
- Technical Accuracy

separately.

Avoid giving the LLM full control over final scoring.

Use LLM outputs as features rather than final decisions.

4.6 Overfitting

Problem

The model may memorize patterns from the training dataset.

Instead of learning interview quality, it learns:

- Specific speaking styles
- Particular candidates
- Dataset-specific characteristics

High training accuracy but poor real-world performance.

Mitigation

Use:

- Cross-validation
- Hold-out test sets
- External datasets

Evaluate unseen candidates.

4.7 Suspicious Behaviour False Positives

Problem

The Suspicious Behaviour Module may incorrectly flag honest candidates.

Example:

Candidate looks away frequently while recalling information.

System interprets:

Looking Away → Reading Script

Impact

The candidate receives an unfair suspicion score.

Mitigation

Use multi-signal detection.

Do not flag candidates based on a single indicator.

Example:

Only trigger high suspicion when:

Looking Away

+

Phone Detected

+

Script Reading Pattern

occur simultaneously.

4.8 Multiple Person Detection Errors

Problem

The system may detect:

- Poster in background
- Reflection in mirror
- Family member walking past

as another participant.

Impact

False cheating alerts.

Mitigation

Track:

- Duration
- Frequency

Ignore short accidental detections.

Only flag when additional faces persist for a significant period.

4.9 Feature Fusion Dominance

Problem

One feature may dominate the final model.

Example:

The model learns that:

Eye Contact = most important

and ignores:

Technical Knowledge

Impact

Technically skilled candidates may be penalized unfairly.

Mitigation

Use:

- Feature importance analysis (SHAP)
- Feature normalization
- Regular retraining

Monitor model behavior continuously.

4.10 Cultural and Personality Differences

Problem

Not all strong candidates behave similarly.

Example:

Some candidates are:

- Introverted
- Quiet
- Soft-spoken

but highly capable.

Impact

The model may incorrectly associate extroversion with competence.

Mitigation

Ensure:

Communication Features $\neq$ Technical Competence

Use separate scoring categories.

Example:

Confidence Score and Technical Knowledge Score should remain independent.

4.11 Data Drift

Problem

Interview styles change over time.

New generations may:

- Use different vocabulary
- Communicate differently
- Interact differently with cameras

Impact

Model performance gradually decreases.

Mitigation

Periodically retrain the model using recent interview data.

Monitor performance metrics continuously.

5.0 Evaluation Methodology

5.1 Evaluation Metrics

To determine whether the proposed AI Interview Assessment System performs effectively, a comprehensive evaluation framework will be implemented. Since the proposed system predicts both continuous interview assessment scores and candidate suitability classifications, both regression and classification metrics will be used.

Regression Metrics

Regression metrics will be used to evaluate predicted interview scores such as Confidence Score, Communication Score, Professionalism Score, Technical Competency Score, and Overall Hireability Score.

Mean Absolute Error (MAE)

MAE measures the average absolute difference between AI-generated scores and ground truth scores assigned by human evaluators. Lower MAE values indicate better prediction accuracy.

Mean Squared Error (MSE)

MSE measures the average squared difference between predicted scores and actual scores. Larger errors receive greater penalties, making MSE useful for identifying substantial prediction mistakes.

Root Mean Squared Error (RMSE)

RMSE is the square root of MSE and provides a more interpretable measure of prediction error because it remains in the same unit as the original score.

R-Squared (R^2)

R^2 evaluates how well the model explains the variance in interview assessment scores. Values closer to 1 indicate stronger predictive performance.

Classification Metrics

If candidates are categorized into classes such as Suitable, Potentially Suitable, and Unsuitable, classification metrics will be used.

Accuracy

Measures the percentage of correctly classified candidates.

Precision

Measures how many candidates predicted as suitable are actually suitable.

Recall

Measures how many truly suitable candidates are correctly identified by the model.

F1-Score

Provides a balanced evaluation by combining Precision and Recall into a single metric.

Spearman ranking

Measures how similarly the AI system and human recruiters rank candidates, regardless of the exact scores assigned.

5.2 Validation Strategy

To ensure the proposed model generalizes effectively to unseen candidates, a robust validation strategy will be employed.

5-Fold Cross Validation

The dataset will be divided into five equal subsets.

For each iteration:

- Four subsets will be used for training.
- One subset will be used for validation.

This process will be repeated five times until every subset has served as the validation set.

The average performance across all folds will be reported.

Advantages

- Reduces overfitting.
- Improves model robustness.
- Provides a reliable estimate of real-world performance.
- Ensures all data samples contribute to both training and validation.

5.3 Testing Procedure

Several experiments will be conducted to evaluate the contribution of different components within the proposed system.

Experiment 1: Transcript-Only Assessment

This baseline model uses:

- Speech-to-Text (Whisper)
- Transcript Evaluation (LLM)

The objective is to evaluate the effectiveness of text-only interview assessment.

Experiment 2: Multimodal Assessment

This model incorporates:

- Video Features

- Audio Features
- Transcript Features

The objective is to determine whether multimodal information improves assessment accuracy compared to transcript-only evaluation.

Experiment 3: Full Proposed System

The complete pipeline will be evaluated using:

- Quality Assessment Module
- Candidate Verification Module
- Video Analysis Module
- Audio Analysis Module
- Transcript Evaluation Module
- Suspicious Behaviour Detection Module
- Feature Fusion Module
- Interview Scoring Module

The results will be compared against Experiments 1 and 2 to determine the effectiveness of the proposed enhancements.

Ablation Study

To measure the contribution of individual modules, selected components will be removed one at a time.

Examples include:

- Without Suspicious Behaviour Detection Module
- Without Candidate Verification Module
- Without Video Features
- Without Audio Features
- Without Feature Fusion Module

Performance differences will indicate the importance of each module to the overall system.

5.4 Human Evaluation Procedure

Interview assessment is inherently subjective. Therefore, human evaluation is required to validate the effectiveness of the AI system.

Step 1: Human Assessment

Recruiters or domain experts will independently assess candidate interviews.

Each evaluator will provide scores for:

- Confidence
- Communication Skills
- Professionalism
- Technical Competency
- Overall Hireability

Step 2: AI Assessment

The proposed AI Interview Assessment System will evaluate the same interviews and generate corresponding scores.

Step 3: Agreement Analysis

The agreement between human evaluators and AI predictions will be measured using:

1. Pearson Correlation

Measures the linear relationship between AI-generated scores and human-generated scores.

2. Spearman Correlation

Measures ranking consistency between AI and human evaluations.

3. Cohen's Kappa

Measures classification agreement between AI predictions and human judgments.

Higher agreement values indicate stronger alignment between AI assessment and human recruiter evaluation.

5.5 Model Comparison Methodology

Multiple machine learning models will be evaluated to determine the most effective scoring model for interview assessment.

Candidate Models

1. Random Forest

Used as a traditional machine learning baseline.

2. XGBoost

Used as the primary model due to its strong performance on structured multimodal feature data.

3. LightGBM

Evaluated for scalability and computational efficiency.

4. Multi-Layer Perceptron (MLP)

Used as a neural network baseline for comparison.

Comparison Criteria

The models will be compared using:

- MAE
- RMSE
- R^2 Score
- Accuracy
- Precision
- Recall
- F1-Score
- Training Time
- Inference Time

The model with the best overall balance between performance and efficiency will be selected for deployment.

5.6 Comparison with Existing Systems

The proposed AI Interview Assessment System is inspired by existing AI interview assessment platforms such as RecruitView and Persuasiv AI. However, the proposed system introduces additional modules focusing on interview integrity, candidate verification, fairness monitoring, and explainable AI.

The comparison below is based on features explicitly described in the RecruitView paper and publicly available information from Persuasiv AI. Features that were not clearly documented are marked as "Not Explicitly Mentioned" rather than assumed to be unavailable.

Capability	RecruitView	Internal Demo System	Persuasiv AI	Proposed System
Multimodal Analysis (Video + Audio + Text)	✓		✓	✓
Transcript-Based Evaluation	✓		✓	✓
Interview Performance Assessment	✓		✓	✓
Personality Assessment	✓		Partial	Optional
Video Behaviour Analysis	✓		Partial	✓
Audio Behaviour Analysis	✓		✓	✓

Candidate Verification	Not Explicitly Mentioned		Not Publicly Mentioned	✓
Multiple Person Detection	Not Explicitly Mentioned		Not Publicly Mentioned	✓
Identity Consistency Verification	Not Explicitly Mentioned		Not Publicly Mentioned	✓
Suspicious Behaviour Detection	Not Explicitly Mentioned		Not Publicly Mentioned	✓
Interview Integrity Assessment	Not Explicitly Mentioned		Not Publicly Mentioned	✓
Feature Fusion	✓		Basic	Advanced
Fairness Monitoring	Limited Discussion		Not Publicly Mentioned	✓
Explainable AI	Limited Discussion		Feedback-Oriented	✓
Human-in-the-Loop Review	Human Labels Used		User Feedback Supported	✓

Compared to existing systems, the proposed framework extends beyond candidate performance assessment by incorporating interview integrity verification, suspicious behaviour detection, fairness monitoring, explainable AI, and human-in-the-loop decision support. These additions aim to improve reliability, transparency, fairness, and trustworthiness in real-world recruitment environments.

5.7 Expected Outcome

The proposed evaluation framework aims to demonstrate that combining video, audio, transcript, and integrity-related features can improve interview assessment performance compared to transcript-only or single-modality approaches.

The proposed system is expected to:

- Improve assessment consistency across candidates.
- Reduce interviewer subjectivity.
- Detect potentially suspicious interview behaviours.
- Improve transparency through explainable AI.
- Support recruiters with objective and data-driven insights.
- Enhance trustworthiness through candidate verification and integrity monitoring.

Rather than replacing human recruiters, the proposed system is designed to function as a decision-support tool that assists recruiters in making more informed hiring decisions.

5.8 Real-World Deployment and Human-in-the-Loop Decision Making

Although AI can automate large portions of the interview assessment process, final hiring decisions should remain under human supervision. Automated systems may occasionally produce incorrect predictions due to dataset bias, transcription errors, cultural differences, unexpected candidate behaviour, or model limitations.

Therefore, the proposed system adopts a Human-in-the-Loop (HITL) approach.

Deployment Workflow

Candidate Interview



AI Assessment Pipeline



Assessment Report Generation



Recruiter Review



Final Hiring Decision

The AI system generates:

- Confidence Score
- Communication Score
- Professionalism Score
- Technical Competency Score
- Hireability Score
- Suspicion Score
- Explainability Report

Recruiters can review these outputs before making final decisions.

High-Risk Cases

The system automatically flags interviews requiring manual review, including:

- High Suspicion Scores
- Multiple Person Detection
- Identity Verification Failure
- Poor Video Quality
- Low Prediction Confidence
- Significant disagreement between AI and human evaluations

For these cases, recruiters may conduct additional verification or secondary interviews before proceeding.

Benefits of Human-in-the-Loop Design

- Prevents overreliance on AI predictions.
- Reduces ethical and legal risks.
- Improves recruiter trust in the system.
- Allows human judgment for exceptional cases.
- Supports responsible deployment of AI in recruitment.

This approach ensures that AI functions as a decision-support tool rather than a replacement for human recruiters.

5.9 Future Improvements

Several future enhancements may further improve the proposed AI Interview Assessment System.

Potential improvements include:

- Real-time interview assessment and feedback.
- Adaptive question generation using Large Language Models. (Qwen 1.7b)
- Multilingual interview support for English, Malay, Mandarin, and other languages.
- Personality assessment using OCEAN personality traits. (DISC test)(HPI-Hogan Personality Inventory)
- Industry-specific interview evaluation models.
- Enhanced fairness monitoring using larger and more diverse datasets.
- Continuous model retraining to address data drift and evolving interview trends.
- Integration with applicant tracking systems (ATS).
- Real-time suspicious behaviour monitoring during live interviews.

These enhancements may further increase system robustness, fairness, scalability, and practical applicability in real-world recruitment environments.

6.0 Conclusion

This proposal presents a comprehensive AI Interview Assessment System that integrates multimodal analysis, candidate verification, suspicious behaviour detection, feature fusion, fairness monitoring, and explainable AI to support objective and reliable candidate evaluation. Unlike traditional interview assessment approaches that rely heavily on human judgment or transcript-only analysis, the proposed system combines video, audio, and textual information to provide a more holistic understanding of candidate performance.

The proposed pipeline begins with Quality Assessment and Candidate Verification to ensure interview recordings are suitable for evaluation and that the correct candidate is being assessed. Video Analysis, Audio Analysis, and Transcript Evaluation modules then extract behavioural, vocal, and semantic features that reflect communication skills, professionalism, confidence, and technical competency. To improve interview integrity, a Suspicious Behaviour Detection Module is introduced to identify potential issues such as multiple-person presence, identity switching, excessive off-screen attention, phone usage, and script reading. These features are subsequently combined through an advanced Feature Fusion Module to generate a comprehensive representation of candidate performance.

Compared with existing systems such as RecruitView and Persuasiv AI, the proposed framework extends beyond performance prediction by incorporating interview integrity assessment, fairness monitoring, explainable AI, and human-in-the-loop decision support. These additions improve transparency, trustworthiness, and practical applicability in real-world recruitment environments.

A comprehensive evaluation framework has also been proposed, including performance metrics, cross-validation strategies, human evaluation procedures, and model comparison methodologies. This ensures that the effectiveness of the proposed system can be systematically assessed and validated. Furthermore, the human-in-the-loop deployment strategy ensures that AI serves as a decision-support tool rather than replacing human recruiters, thereby reducing ethical and legal risks associated with automated hiring systems.

In conclusion, the proposed AI Interview Assessment System aims to improve recruitment efficiency, enhance assessment consistency, reduce interviewer subjectivity, and strengthen interview integrity. By combining multimodal analysis with explainable and responsible AI practices, the system has the potential to provide more reliable, fair, and transparent candidate evaluations while supporting recruiters in making informed hiring decisions.